

GEOG7305 Programming for Geospatial Data

Project Guideline

For the project, students are expected to form a small group (**less than 4 members**) and each group should complete an independent GIS project. The project topic can be related to any implementation and customization of GIS on a relatively complex geographical problem.

The project will be identified by the students and should be approved by the instructor before the 6th week (**6 March, 2026**). A written ***project plan*** is required within the first eight weeks of the semester and the students are required to present the plan on **20 March, 2026**. This should clearly describe: (i) the scope and objective of the project; (ii) the data to be used and their sources; (iii) the processing, programming, and/or analysis to be applied to the data in order to meet the objective; and (iv) the results and conclusions you expect to obtain.

Students are required to present their project and write up a project report. The project presentation will be held at the end of the semester and the project report will be due in one week after the Final Examination of this course (**8 May, 2026**).

The ***project deliverables*** at the semester end must include:

- (1) a soft-copy report documenting the project;
 - (2) power point files for presentations;
 - (3) any map, computer program, ArcGIS script or database resulting from the project.
- The deliverables should be submitted to a folder in Moodle that will be specified by the instructor.

Group presentations:

- Mid-term on 8th week (20 March, 2026)
- Last teaching week (24 April, 2026)
- 10 minutes each group

Project report:

- Group report
- ≤20 A4 pages (1.5 lines space)
- Font: Times New Roman 12
- Margins: top/bottom 2 cm, left/right 2cm

Report content:

- Introduction
- User Requirement Analysis (e.g., Purpose and Description; Functions Required; Data Required, etc.)
- System Design (e.g., Conceptual Design; Database Requirement and Analysis; Data Flow; User Interface Design; System Architecture Design, etc.)
- Project Development Procedures (e.g., Data Collection and Processing; Function Exploration and Processing; Testing, etc.)

- Result Interpretation - Case study
- Evaluation (e.g., limitation; strength, significance, future prospect)

Important Dates

- Propose research topic (6 March, 2026)
- Mid-term Presentation & Proposal (20 March, 2026)
- Final Presentation (24 April, 2026)
- Report Due: (8 May, 2026)

Evaluation

- By instructor and student groups
- Criteria
 - Significance of application (20%)
 - Technical quality (60%)
 - Presentation (20%)